

Magnetic field & Magnetic Forces:

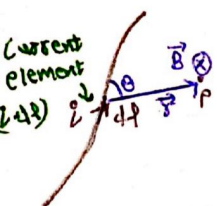
1) Static charges → produces only EF & only EF can exert force on these

2) Moving charges → Produces both EF & MF and both fields can exert force on these

3) Current carrying Conductor (Uncharged) → Produces only MF & only MF can exert force on moving charges inside conductor

Biot Savart's law:

→ Tells us expression for MF at any point due to current carrying wire.



$$dB = \frac{\mu_0}{4\pi} \frac{I \cdot dl \sin\theta}{r^2}$$

$$d\vec{B} = \frac{\mu_0}{4\pi} \frac{I (d\vec{l} \times \vec{r})}{r^3}$$

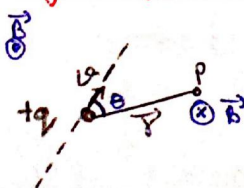
Dir'n of $\vec{B}$ is given by right hand thumb rule.

→ Put right thumb along $d\vec{l}$ & fingers along $\vec{r}$, then curling fingers give $\vec{B}$.

Ex: Find $\vec{B}$ at O due to small current element. Since here, instead of point's position vector, we are give current element's position vector. We put a (-) sign

$$\vec{B} = -\frac{\mu_0}{4\pi} \frac{I dl \cos\alpha}{r^2} (-\hat{r}) \quad \text{Ans}$$

Magnetic field due to moving charges:



$$B = \frac{\mu_0}{4\pi} \frac{qv \sin\theta}{r^2}$$

$$\vec{B} = \frac{\mu_0}{4\pi} \frac{q(\vec{v} \times \vec{r})}{r^3}$$

→ Dir'n given by keeping right thumb along $\vec{v}$ & fingers along $\vec{r}$, then curling gives $\vec{B}$.

→ Equivalent instantaneous equivalent element of a moving charge is

$$[Idl] = qv$$

NOTE: μ_0 = Permeability of free space
 $= 4\pi \times 10^{-7} \text{ T.m/A}$

Magnetic field due to Current Carrying wire:

Wire: (Infinite wire) (Semi-Infinite wire)



$$B = \frac{\mu_0 I}{4\pi r} (\cos\theta_1 + \cos\theta_2)$$

$$B = \frac{\mu_0 I}{2\pi r}$$

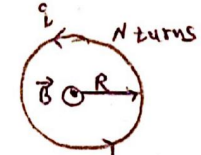
$$B = \frac{\mu_0 I}{4\pi r}$$

Magnetic field due to Circular Coil:

For any shaped coil, MF inside and outside are in opp dir'n. As they form closed loops.

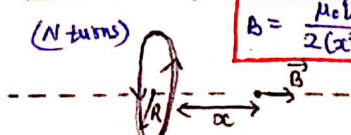
→ For dir'n of $\vec{B}$, curl your right hand finger along flow of current, then thumb gives dir'n of $\vec{B}$.

1) $\vec{B}$ at centre of coil →

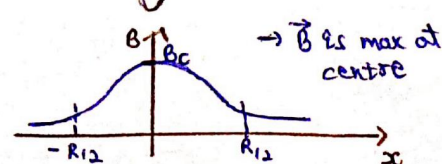


$$B = \frac{\mu_0 I}{2R} \times N$$

2) $\vec{B}$ at axis of coil →



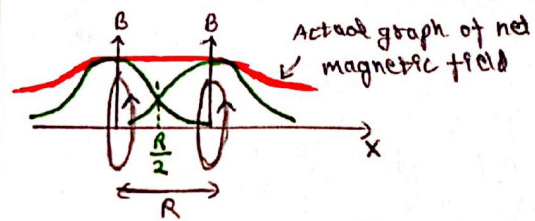
$$B = \frac{\mu_0 I N^2}{2(x^2 + R^2)^{3/2}} \times N$$



→ After $R/2$ distance, graph shifts from concave to convex in nature.

Helmholtz Coils →

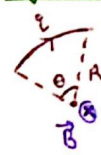
→ Pair of similar coils separated by R (their radius) which produce almost uniform MF in b/w them.



Put $x=R/2$ in expression of MF

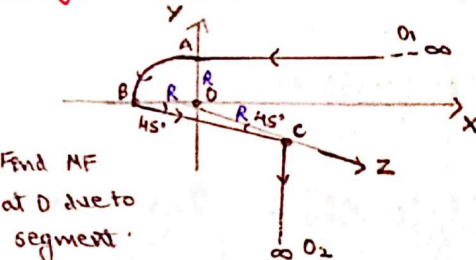
$$B_{OF} = \frac{8}{5\sqrt{5}} \frac{\mu_0 I N}{R}$$

Magnetic field at centre of a circular Arc:



$$B = \frac{\mu_0 I \theta}{4\pi R}$$

Magnetic Field due to 3D wire segment:



Find MF at P due to segment.

$$d\vec{B} = \vec{B}_{OA} + \vec{B}_{AB} + \vec{B}_{BC} + \vec{B}_{CO}$$

$$\vec{B}_O = \frac{\mu_0 I}{4\pi R} \hat{r} + \frac{\mu_0 I}{8R} \hat{r} + \frac{\mu_0 I}{4\pi (\frac{R}{\sqrt{2}})} \hat{r} + \frac{\mu_0 I}{4\pi R} \hat{r}$$

Magnetic Field due to Current Carrying:

Strip: Consider a small element dx



$$dI = \frac{I}{b} \cdot dx$$

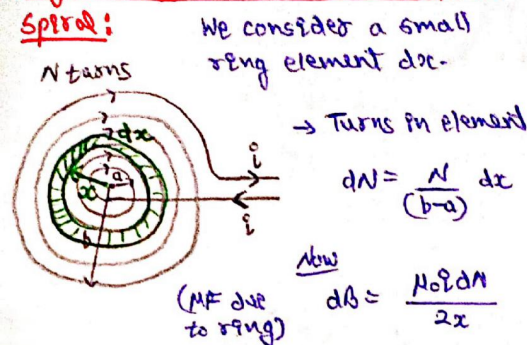
At P

$$dB = \frac{\mu_0 \cdot dI}{2\pi x}$$

$$B_p = \int dB = \frac{\mu_0 I}{2\pi b} \int \frac{dx}{x}$$

$$\Rightarrow B_p = \frac{\mu_0 I}{2\pi b} \ln\left(\frac{b+x}{x}\right)$$

Magnetic Field due to Current Carrying Spiral:



$$\Rightarrow B_0 = \int dB = \frac{\mu_0 i N}{2(b-a)} \int_a^b \frac{dx}{x}$$

$$\Rightarrow B_0 = \frac{\mu_0 i N}{2(b-a)} \ln\left(\frac{b}{a}\right)$$

(MF due to entire spiral at its centre)

Magnetic Field due to a Solenoid:

$\rightarrow$ Solenoid is tightly coiled, radii of loops is very small, length of solenoid is long.

$\rightarrow$ Outside solenoid, $\vec{B} = 0$
Inside solenoid, $\vec{B}$ is uniform at axis ($n \rightarrow$ Turns/Length)

1) $B_p = \mu_0 n i$

(∞ -solenoid)

2) $B_{end} = \frac{1}{2} \mu_0 n i$

(Semi- ∞ solenoid) (here, point P is at end)

3) $B_p = \frac{1}{2} \mu_0 n i (\cos\theta_1 + \cos\theta_2)$

(finite solenoid)

Magnetic Field due to a Toroid:

(N turns)

$$B_{in} = \frac{\mu_0 N i}{2\pi R}$$

($N \rightarrow$ Total no. of turns)

($R \gg r$)

Magnetic Field due to large Current Carrying Sheet:

1) Thin Sheet $\Rightarrow$

(up) $\Rightarrow B = \frac{1}{2} \mu_0 i$

(down) $\Rightarrow B = -\frac{1}{2} \mu_0 i$

Above is side view of sheet, showing thickness.

$\rightarrow$ Put right thumb along current, so curling fingers give $\vec{B}$.

2) Thick Sheet $\Rightarrow$

$\rightarrow B_{out} = \frac{1}{2} \mu_0 J d$ $B_{in} = \mu_0 J x$

(side view of sheet showing thickness)

$[J \rightarrow$ current density]

Ampere's Law:

① i_1 ② i_2 ③ i_3 ④ i_4

$\rightarrow$ Outward current = +ve
Inward current = -ve

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{enc.}$$

$$\Rightarrow \oint \vec{B} \cdot d\vec{l} = \mu_0 (i_1 + i_2 - i_3)$$

MF due to Current carrying cylinder:

(Hollow Cylinder) (Solid Cylinder)

(Hollow Cylinder)

$B_{in} = 0$

$B_{out} = \frac{\mu_0 i}{2\pi r}$

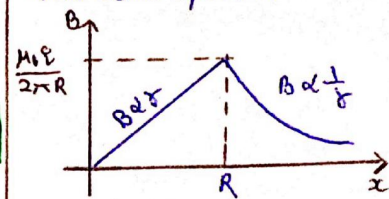
(Solid Cylinder)

$B_{out} = \frac{\mu_0 i}{2\pi r}$

$B_{in} = \frac{\mu_0 i x}{2\pi R^2}$

$B \propto \frac{1}{r}$

$\rightarrow$ For solid cylinder



MF due to Revolving charge:

$\rightarrow$ For a revolving charge, we can write its equivalent current in orbit due to its revolution as

$$i = \text{charge} \times \text{Rev. freq. } (f)$$

$$\Rightarrow i = q \cdot \frac{\omega}{2\pi} = \frac{q\omega}{4\pi R^2}$$

$\rightarrow$ We can use this eq. current density, only for $\vec{B}$ calculation at centre. Not applicable everywhere

$B_0 = \frac{\mu_0 i}{2R}$

$\Rightarrow B_0 = \frac{\mu_0 q \omega}{4\pi R^2}$

$B_p = \frac{\mu_0}{4\pi} \frac{q\omega}{(R^2 + x^2)^{3/2}}$

$(B_p)_{\text{Along axis}} = \frac{\mu_0 q \omega R}{4\pi R (R^2 + x^2)^{3/2}}$

MF due to Rotating Disc:

$dq = \sigma \cdot 2\pi x \cdot dx$

$dI = dq \cdot \frac{\omega}{2\pi}$

$\Rightarrow B_0 = \int \frac{\mu_0 dI}{2x} = \int_0^R \frac{\mu_0 (\frac{\omega}{2\pi}) \sigma 2\pi x dx}{2x}$

$B_0 = \frac{\mu_0 \omega R \sigma}{2}$ Ans

MF due to Coaxial Cable:

$I_{enc} = (I - \frac{I}{(b^2 - a^2)} \times \pi(x^2 - a^2))$

$B_{out} = 0$

$\Rightarrow I_{enc} = I - \frac{I}{(b^2 - a^2)} (x^2 - a^2)$

$a < x < b$ $x < a$ $b < x < c$

$B = \frac{\mu_0 I x}{2\pi x^2}$ $B = \frac{\mu_0 I x}{2\pi a^2}$ $B = \frac{\mu_0 I_{enc}}{2\pi x}$